Class 8: Breast Cancer Mini Project

Anisa Mody (PID: A19145291)

Table of contents

Background 1
Data Import 1
Exploratory Data Analysis 3
Principal Component Analysis (PCA) 4
Hierarchical Clustering 11
Combining Methods 14
Prediction 17

Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

Today, we will analyze a biopsy data set from a fine needle aspiration (FNA) of a breast mass that describe characteristics of the cell nuclei present in digitized images.

Data Import

This data is made available as a CSV file for download. We can read this in using `read.csv()`:

```
# Read in the Wisconsin breast cancer data from the CSV file
fna.data <- "WisconsinCancer.csv"

# Load the CSV file into R, using the first column as row names
wisc.df <- read.csv(fna.data, row.names=1)

# Preview the first 3 rows of the dataset
head(wisc.df, 3)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38          122.8      1001
842517           M       20.57        17.77          132.9      1326
84300903         M       19.69        21.25          130.0      1203
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
         perimeter_worst area_worst smoothness_worst compactness_worst
842302             184.6       2019           0.1622            0.6656
842517             158.8       1956           0.1238            0.1866
84300903           152.5       1709           0.1444            0.4245
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
```

Make sure we remove or exclude the `diagnosis` column from the data set that we use for further analysis - this is the expert diagnosis as either M or B:

```r
# Remove the diagnosis column so PCA only uses numeric measurement variables
wisc.data <- wisc.df[,-1]

# Save the diagnosis labels as a separate factor variable
diagnosis <- as.factor(wisc.df$diagnosis)
```

Exploratory Data Analysis

Question 1. How many observations are in this dataset?

```r
# Count the number of observations in the dataset
nrow(wisc.data)
```

```
[1] 569
```

Question 2. How many of the observations have a malignant diagosis?

```r
# Count how many samples are benign or malignant
table(wisc.df$diagnosis)
```

```

  B   M 
357 212 
```

Question 3. How many variables/features in this data are suffixed with `_mean`?

We can use the `grep()` function to help us here:

```r
# Show all column names in the numeric cancer dataset
colnames(wisc.data)
```

```
 [1] "radius_mean"             "texture_mean"           
 [3] "perimeter_mean"          "area_mean"              
 [5] "smoothness_mean"         "compactness_mean"       
 [7] "concavity_mean"          "concave.points_mean"    
 [9] "symmetry_mean"           "fractal_dimension_mean" 
[11] "radius_se"               "texture_se"             
[13] "perimeter_se"            "area_se"                
[15] "smoothness_se"           "compactness_se"         
[17] "concavity_se"            "concave.points_se"      
[19] "symmetry_se"             "fractal_dimension_se"   
[21] "radius_worst"            "texture_worst"          
[23] "perimeter_worst"         "area_worst"             
[25] "smoothness_worst"        "compactness_worst"      
[27] "concavity_worst"         "concave.points_worst"   
[29] "symmetry_worst"          "fractal_dimension_worst"
```

```r
# Count how many variables end with "_mean"
length(grep("_mean", colnames(wisc.data), value = T))
```

```
[1] 10
```

Principal Component Analysis (PCA)

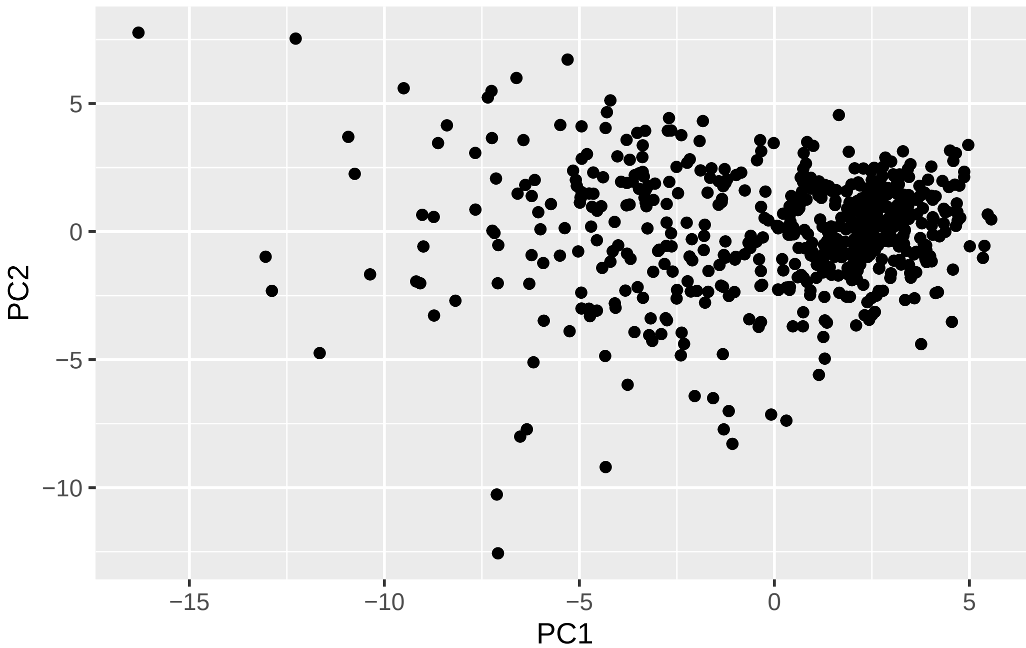
We need to scale our data before PCA with the `scale=TRUE` argument to `prcomp()`.

```r
# Run PCA on the numeric data and scale variables first
wisc.pr <- prcomp(wisc.data, scale=TRUE)

# Display a summary of variance explained by each principal component
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

```r
# Loads ggplot2 for plotting
library(ggplot2)
```

```r
#Plot PC1 against PC2
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point()
```



Question 4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

```r
# Calculate the variance for each principal component
summary(wisc.pr)$importance[2, 1]
```

```
[1] 0.44272
```

Question 5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

```r
which(summary(wisc.pr)$importance[3, ] >= 0.70)[1]
```

```
PC3
  3
```

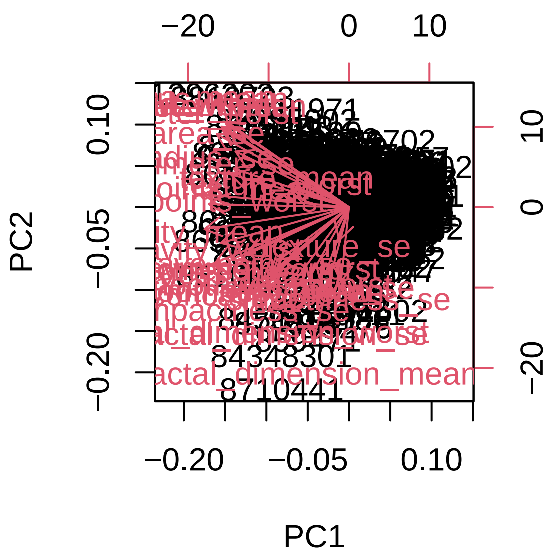
Question 6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

```r
which(summary(wisc.pr)$importance[3, ] >= 0.90)[1]
```

```
PC7
  7
```

Question 7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

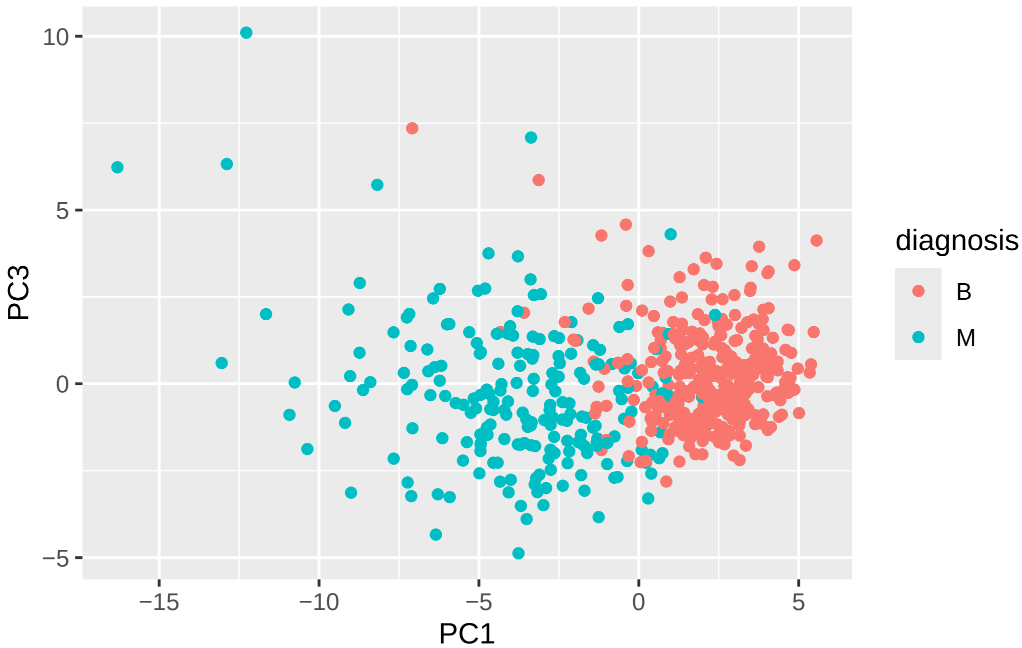
```r
biplot(wisc.pr)
```



The plot is very cluttered, with many overlapping points and variable arrows, making it hard to read. It's difficult to clearly distinguish patterns or individual observations because of the excessive labeling. While it contains useful information, the overall complexity makes it challenging to interpret quickly.

Question 8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```r
# Plot PC1 against PC2
ggplot(wisc.pr$x) +
  aes(PC1, PC3, col = diagnosis) +
  geom_point()
```



The PC1 vs PC2 plot typically shows a clearer separation between diagnosis groups, while the PC1 vs PC3 plot shows less distinct separation. This indicates that PC2 captures more of the variation relevant to distinguishing the groups than PC3 does.

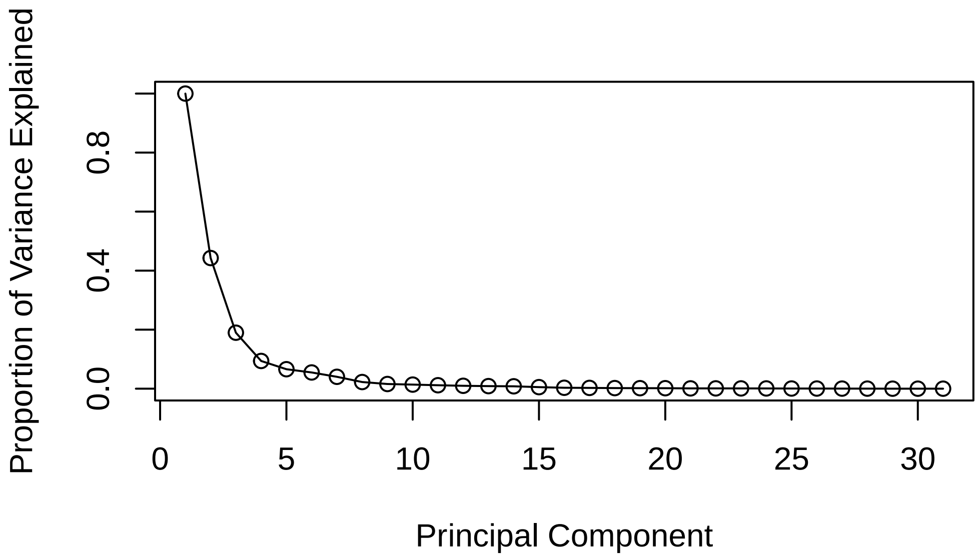
Variance

```r
# Calculate the variance for each principal component
pr.var <- wisc.pr$sdev^2

#View the first few variance values
head(pr.var)
```

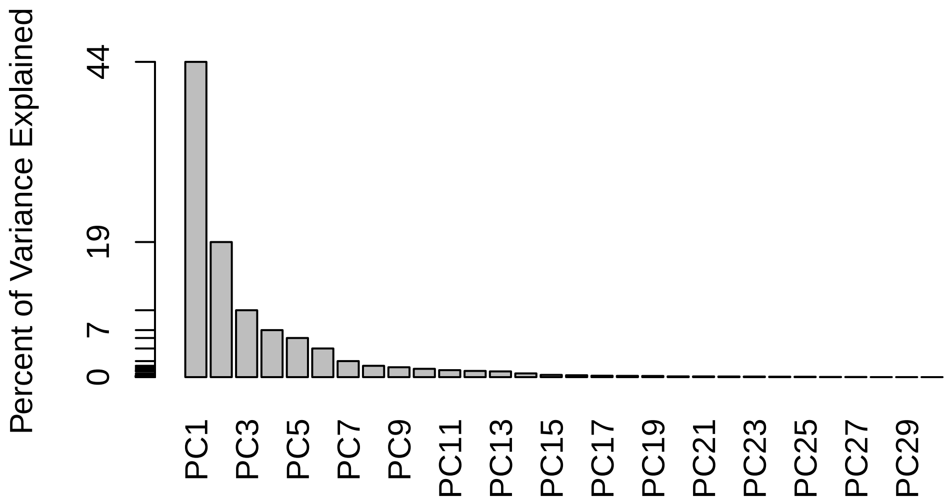
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```r
# Variance explained by each principal component: pve
pve <- pr.var / sum(pr.var)
```

```r
# Plot variance explained for each principal component
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```



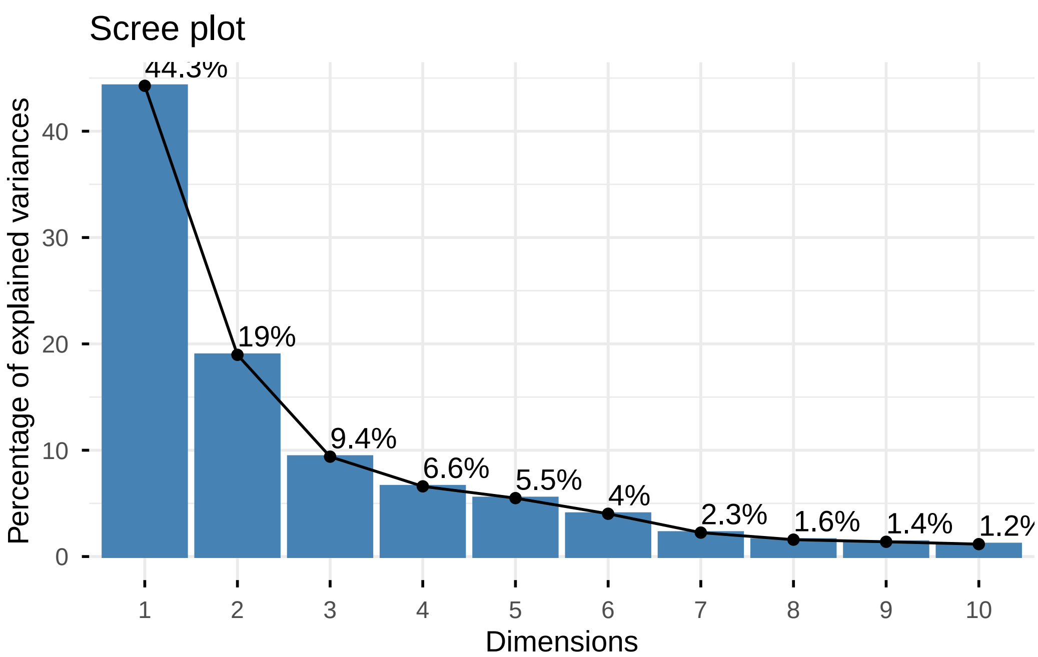

```r
# Alternative scree plot of the same data, note data driven y-axis
barplot(pve, ylab = "Percent of Variance Explained",
     names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```




```
## ggplot based graph
#install.packages("factoextra")
library(factoextra)
```

```
Welcome to factoextra!

Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/product/
guide-to-principal-component-methods-in-r/
```

```
fviz_eig(wisc.pr, addlabels = TRUE)
```



Question 9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation["concave.points_mean", 1]
```

```
[1] -0.2608538
```

```
# Sort loadings by magnitude (largest first)
sort(abs(wisc.pr$rotation[,1]), decreasing = TRUE)[1:5]
```

```
 concave.points_mean       concavity_mean concave.points_worst 
           0.2608538            0.2584005            0.2508860 
    compactness_mean      perimeter_worst 
           0.2392854            0.2366397 
```

```
sort(abs(wisc.pr$rotation[,1]), decreasing = TRUE)
```

```
    concave.points_mean          concavity_mean    concave.points_worst 
             0.26085376              0.25840048              0.25088597 
```

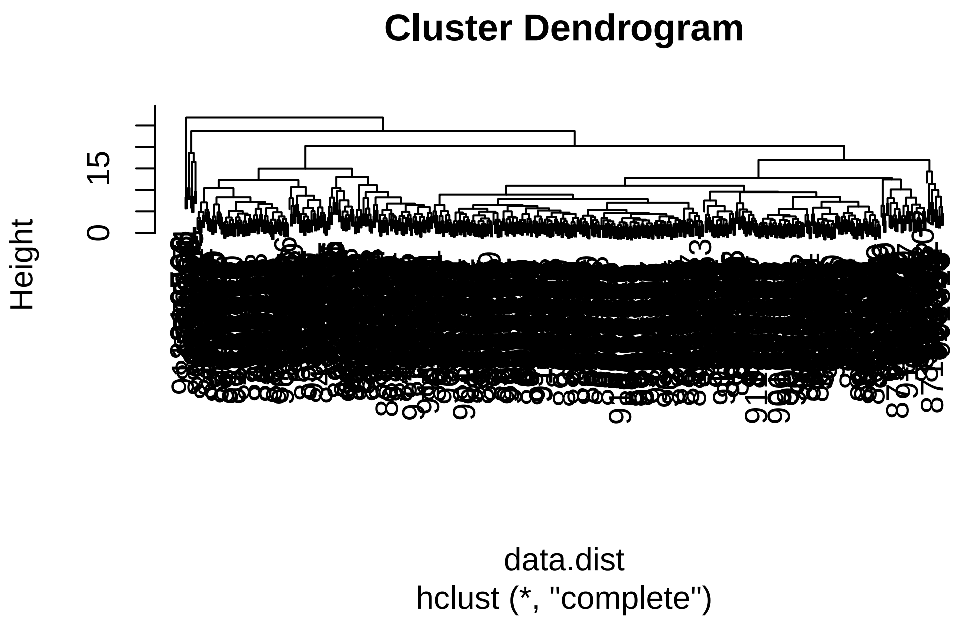
```
   compactness_mean         perimeter_worst          concavity_worst
         0.23928535              0.23663968               0.22876753
       radius_worst         perimeter_mean               area_worst
         0.22799663              0.22753729               0.22487053
          area_mean             radius_mean             perimeter_se
         0.22099499              0.21890244               0.21132592
  compactness_worst               radius_se                  area_se
         0.21009588              0.20597878               0.20286964
  concave.points_se          compactness_se             concavity_se
         0.18341740              0.17039345               0.15358979
    smoothness_mean           symmetry_mean  fractal_dimension_worst
         0.14258969              0.13816696               0.13178394
   smoothness_worst          symmetry_worst            texture_worst
         0.12795256              0.12290456               0.10446933
       texture_mean    fractal_dimension_se   fractal_dimension_mean
         0.10372458              0.10256832               0.06436335
        symmetry_se              texture_se            smoothness_se
         0.04249842              0.01742803               0.01453145
```

The loading for concave.points_mean on PC1 can be obtained with `wisc.pr$rotation["concave.points_mean", 1]`, and it has a relatively large value, indicating it strongly contributes to PC1. However, when comparing all loadings, there are a few features with even larger absolute values. This means that while it is important, it is not the single strongest contributor to PC1.

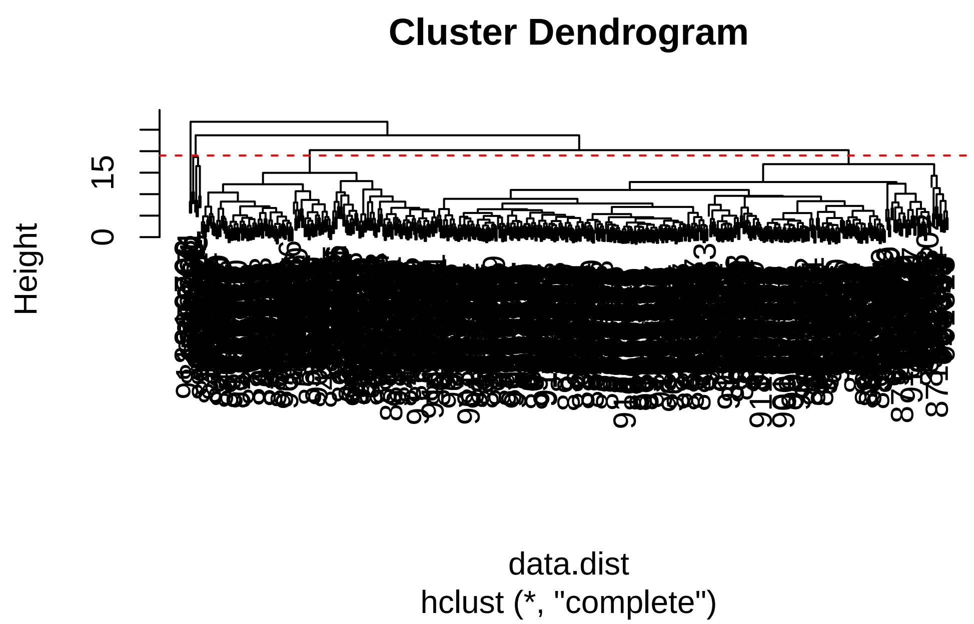
Hierarchical Clustering

> Question 10. Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters?

```r
# Scale the wisc.data data using the "scale()" function
data.scaled <- scale(wisc.data)

# Calculate Euclidean distances between all samples
data.dist <- dist(data.scaled)

# Perform hierarchical clustering using complete linkage
wisc.hclust <- hclust(data.dist, method = "complete")

# Plot the clustering dendrogram
plot(wisc.hclust)
```




```r
# Scale the wisc.data data using the "scale()" function
data.scaled <- scale(wisc.data)

# Calculate Euclidean distances between all samples
data.dist <- dist(data.scaled)

# Perform hierarchical clustering using complete linkage
wisc.hclust <- hclust(data.dist, method = "complete")

# Plot the clustering dendrogram
plot(wisc.hclust)

# Add a horizontal line around height 19, which gives about 4 clusters
abline(h = 19, col = "red", lty = 2)
```



Question 12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

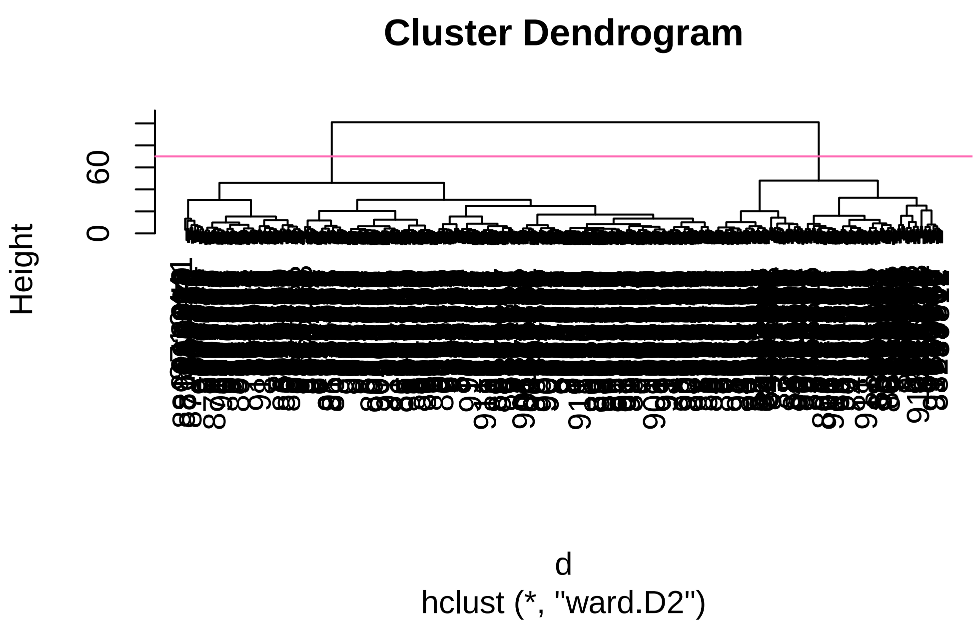
```
# Cluster using the first 7 principal components, which explain over 90% variance
wisc.pr.hclust <- hclust(dist(wisc.pr$x[, 1:7]), method = "ward.D2")

# Cut the PCA-based dendrogram into 2 clusters
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k = 2)

# Compare PCA-based clusters to the expert diagnosis
table(wisc.pr.hclust.clusters, diagnosis)
```

```
                       diagnosis
wisc.pr.hclust.clusters   B   M
                      1  28 188
                      2 329  24
```

I prefer ward.D2 because it minimizes within-cluster variance, which produces tighter and more compact clusters. In this dataset, it results in clearer separation between benign and malignant samples compared to complete linkage. Complete linkage usually forms more irregular clusters and shows weaker alignment with the known diagnoses.

Combining Methods

```r
# Calculate the distance matrix using the first four principal components
d <- dist(wisc.pr$x[,1:4])

# Run hierarchical clustering on the PCA distance matrix using Ward's method
wisc.pr.hclust <- hclust(d, method = "ward.D2")

# Plot the hierarchical clustering dendrogram
plot(wisc.pr.hclust)

#Add a horizontal line at height 70 to show where the tree can be cut into groups
abline(h=70, col="hotpink")
```




```r
grps <- cutree(wisc.pr.hclust, h=70)
table(grps)
```

```
grps
  1   2 
171 398 
```

How does this clustering `grps` correspond to the expert `diagnosis`?

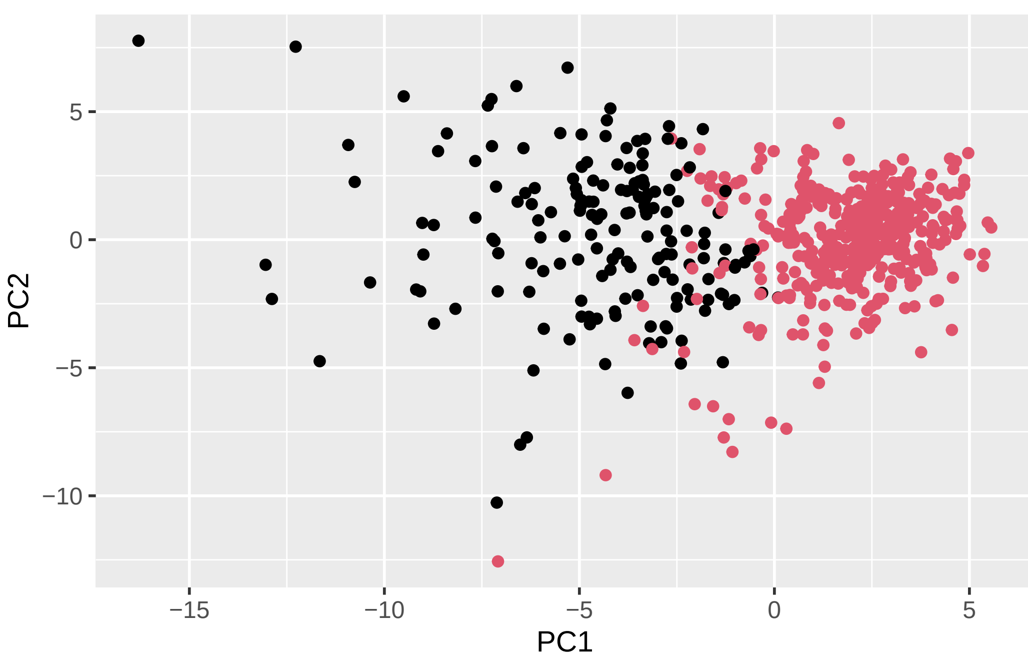
```
table(diagnosis)
```

```
diagnosis
  B   M
357 212
```

```
table(diagnosis, grps)
```

```
         grps
diagnosis   1   2
        B   6 351
        M 165  47
```

```
# Plot PC1 against PC2 with the points colored based on a vector `grps`
ggplot(wisc.pr$x) +
aes(PC1, PC2) +
geom_point(col=grps)
```



Clustering on PCA Results

Question 13. How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses?

```
# Compare the original hierarchical clustering result to diagnosis
table(wisc.pr.hclust.clusters, diagnosis)
```

```
                       diagnosis
wisc.pr.hclust.clusters   B   M
                      1  28 188
                      2 329  24
```

Question 14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

```
# Create clusters from the scaled-data hierarchical clustering model
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)

# Compare the non-PCA hierarchical clustering groups to the actual diagnoses
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

```
# Compare the PCA-based clustering groups to the actual diagnoses again
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1   6 165
   2 351  47
```

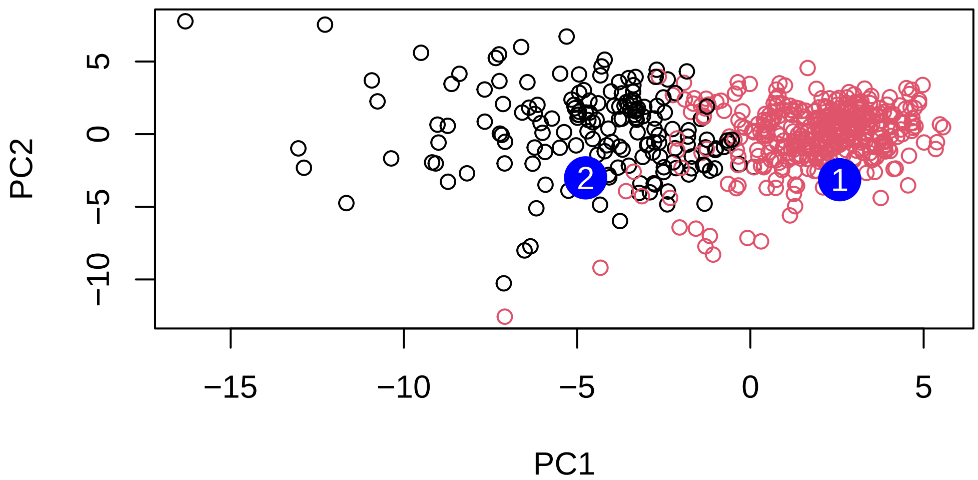
The hierarchical clustering without PCA separates the diagnoses less effectively, with several clusters containing a mix of both benign and malignant samples. In contrast, the PCA-based clustering shows clearer grouping, with clusters more strongly dominated by one diagnosis type. This suggests that using PCA before clustering improves separation by reducing noise and focusing on the most informative variation in the data.

Prediction

```r
# Save the URL for the new patient samples
url <- "https://tinyurl.com/new-samples-CSV"

# Read the new patient samples into R
new <- read.csv(url)

# Project the new samples into the existing PCA model
npc <- predict(wisc.pr, newdata=new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```r
# Plot original samples using PC1 and PC2, colored by cluster group
plot(wisc.pr$x[,1:2], col=grps)

# Add the new samples as large blue points
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)

# Label the new samples as 1 and 2
text(npc[,1], npc[,2], c(1,2), col="white")
```



Question 16. Which of these new patients should we prioritize for follow up based on your results?

Sample 2 should be prioritized for follow-up because it lies within or closest to the cluster of malignant samples in the PCA plot. This suggests it shares similar characteristics with known cancerous cases. In contrast, sample 1 appears closer to the benign cluster and is therefore less concerning.